Sensitivity to the reference configuration. We fixed one UP sample in each of three settings: Clustered $(N, n) = (144, 16)$, Meuse $(155, 10)$, and Swiss $(959, 100)$. Procedure A varied TSP search while fixing the OT coupling and preliminary labels; B varied numerical OT routes under a deterministic TSP rule; C reused the corresponding B coupling with independent TSP seeds. We retained 40 outcomes per procedure and setting. All 360 saved reference configurations and translations were reused, without repeating reference optimization. We recalculated only the densities and DI, using a spherical uniform kernel and scalar Scott bandwidth $h = N^{-1/(d+4)}$ in the original normalized coordinate system. The same bandwidth was used after translation, without renormalizing the translated coordinates. This is a KDE-only comparison on the saved references.

Table 1 summarizes the results. The baseline DIs were 0.276619, −0.275822, and −0.085989, respectively. All signs agreed with the corresponding baseline. Reference invariance was nevertheless not observed: in the Clustered setting, two outcomes sharing exactly the OT coupling and preliminary labels gave DI 0.276619 and 0.283761, an absolute difference of 0.007141. Their open-TSP lengths were within 1.892% of the saved exact benchmark for that common centroid instance. Their native and common-scale OT gaps relative to the best observed saved values were 1.042% and 0.617%, respectively; hence this comparison satisfies the 2% screens, but not the 1% screen. Swiss variation includes ambiguity in numerical hardening near coincident centroids and should not all be interpreted as variation between distinct mathematical OT optima.

The historical-KDE rescoring reduced the observed spread and changed DI magnitudes substantially relative to the previous box-KDE calculation. These conditional, dependent search outcomes quantify a limitation of the reference construction; they do not establish general reference invariance, population-level frequencies, sampling uncertainty, or agreement with the manuscript's Monte Carlo averages.

Table 1: Fixed-sample reference sensitivity after historical-KDE rescoring: Clustered $(N, n) = (144, 16)$, Meuse $(155, 10)$, and Swiss $(959, 100)$. Each row contains 40 saved evaluations; 360 in total, with three separate baseline controls. DI_0 uses spherical top-hat KDE with scalar Scott bandwidth. SD uses denominator 39. Max denotes $\max |DI - DI_0|$; flips compare signs with the corresponding baseline. Box SD refers to the previous covariance-scaled box KDE on exactly the same references. These are descriptive search summaries, not sampling uncertainty.

Population	Proc.	DI_0	DI range	SD(box)	SD(radial)	Max	Flips
Clustered	A	0.276619	[0.276619,0.293681]	0.008879	0.003794	0.017062	0
Clustered	B	0.276619	[0.274079,0.276619]	0.001324	0.001227	0.002540	0
Clustered	C	0.276619	[0.274026,0.283761]	0.004220	0.002879	0.007141	0
Meuse	A	-0.275822	[-0.275822,-0.269310]	0.004421	0.001030	0.006512	0
Meuse	B	-0.275822	[-0.275822,-0.275822]	0.000000	0.000000	0.000000	0
Meuse	C	-0.275822	[-0.275822,-0.269310]	0.004421	0.001030	0.006512	0
Swiss	A	-0.085989	[-0.106772,-0.070290]	0.022235	0.008029	0.020783	0
Swiss	B	-0.085989	[-0.087118,-0.075647]	0.011025	0.002309	0.010342	0
Swiss	C	-0.085989	[-0.094809,-0.064337]	0.019282	0.006475	0.021651	0